

PAPER_965: NS Phonon Effects for GW190425

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Abstract

We derive neutron star phonon effects for GW190425 (mass-gap BNS merger). The UQFF phonon-corrected strain $h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot 0.5297 \cdot \exp([SSq] \cdot t/26)$ produces a 47% strain reduction. The phonon-corrected tidal deformability $\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}}(1 + \delta\Lambda)$ matches LIGO constraints. Mass-gap probability: P(NS) = 49%, P(BH) = 51%.

1. Phonon-Corrected Strain

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot 0.5297 \cdot \exp\left(\frac{[SSq] \cdot t}{26}\right)$$

2. Wavelength Correction

$$\lambda_{\text{UQFF}} = \lambda_{\text{GR}} \cdot \left(1 - \frac{F_{UBi}}{F_U} \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)\right)$$

3. Tidal Deformability

$$\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} \cdot (1 + \delta\Lambda_{\text{phonon}})$$

4. Mass-Gap Classification

At $\Gamma = 0.1$ THz: P(NS) $\approx$ 49%, P(BH) $\approx$ 51%.

References

1. LIGO/Virgo - GW190425 (2020)
2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)